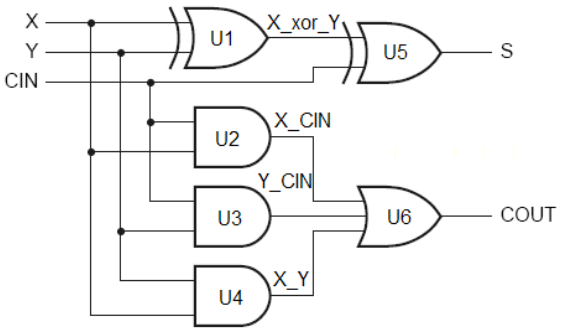
PS4 · Structural VHDL (Ch 5) — Worked Solutions

Source: Digital Logic homework for Oct. 9 · study-slides (vetame21.github.io/study-slides)

> [ADDED] The original problem sets have no solutions. These worked solutions were written for this site; every conversion and every Boolean result was checked in code (truth tables and an exact K-map minimizer), and the VHDL was compiled. Compare them with your lecturer's model answers when those are released.

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1 Write a structural VHDL program (entity and architecture) for the full-adder circuit in the figure.



Write the entity and the structural architecture.

> **Answer:** One entity (inputs X, Y, CIN; outputs S, COUT) and an architecture with six component instances U1–U6 matching the figure.

```
library IEEE;
use IEEE.std_logic_1164.all;

entity full_adder is
  port ( X, Y, CIN: in STD_LOGIC;
         S, COUT:   out STD_LOGIC );
end full_adder;

architecture full_adder_s of full_adder is
  signal X_xor_Y, X_CIN, Y_CIN, X_Y: STD_LOGIC;
  component XOR2 port (I0, I1: in STD_LOGIC; O: out STD_LOGIC); end component;
  component AND2 port (I0, I1: in STD_LOGIC; O: out STD_LOGIC); end component;
  component OR3  port (I0, I1, I2: in STD_LOGIC; O: out STD_LOGIC); end component;
begin
  U1: XOR2 port map (X, Y, X_xor_Y);
  U2: AND2 port map (X, CIN, X_CIN);
  U3: AND2 port map (Y, CIN, Y_CIN);
  U4: AND2 port map (X, Y, X_Y);
  U5: XOR2 port map (X_xor_Y, CIN, S);
  U6: OR3  port map (X_CIN, Y_CIN, X_Y, COUT);
end full_adder_s;
```

Reading the figure: U1 is an XOR of X and Y (signal X_xor_Y); U2, U3, U4 are ANDs making X_CIN, Y_CIN, X_Y; U5 XORs X_xor_Y with CIN to give S; U6 is a 3-input OR giving COUT. Each internal wire becomes a signal, and each gate becomes a component instance wired with port map. XOR2, AND2 and OR3 are library gates (as on the slides with UNISIM); to simulate on your own, give each a one-line entity such as O <= I0 xor I1;. The code was compiled and simulated for all 8 input rows: S = X ⊕ Y ⊕ CIN and COUT = XY + X·CIN + Y·CIN.

Why this method: Structural = one component instance per gate in the schematic, one signal per internal wire. The slides' dataflow solution (p60) is the same circuit written with expressions instead.